TEXORA · LECTURE SHEET

Colour Chemistry: Reactive, Disperse, Azoic, Sulphur Dyes and Pigments

A dye molecule is a chromophore fitted with a delivery system. This sheet reads the five great application classes of this course by their chemistry — how the colour is built, how it reaches the fibre, and how it is made to stay.

WP-4101 · TEXTILE CHEMISTRY & COLORATION-III · TOPIC 2 · SPECIALIZATION — WET PROCESSING

1. The anatomy of a dye molecule

Colour arises in the **chromophore** — a conjugated π-electron system (azo −N=N−, anthraquinone, phthalocyanine, methine) whose electronic transitions absorb visible light. **Auxochromes** (−NH_2, −OH, and their substituted forms) deepen and shift the colour and often provide the fibre attachment. Around this coloured core, each application class builds its own machinery: solubilising groups (−SO_3Na) for water-borne dyes, reactive groups for covalent fixation, or deliberate insolubility for disperse dyes and pigments. Substituent effects follow rules worth remembering: electron-donating groups shift absorption to longer wavelengths (bathochromic, deeper colour); electron-withdrawing groups do the reverse (hypsochromic); extending conjugation deepens colour and usually raises substantivity.

2. Reactive dyes — colour by covalent bond

Reactive dyes carry a group that reacts chemically with the fibre: with cellulose hydroxyls under alkaline conditions, or with amino groups of wool, silk and nylon. Two reaction families dominate:

- **Nucleophilic substitution** — chlorotriazine systems (dichlorotriazine: cold-reacting; monochlorotriazine: hot-reacting): cellulose−O^- displaces chlorine on the triazine ring, forming a covalent ether bond dye–fibre.
- **Nucleophilic addition** — vinyl sulfone systems (applied as the masked β-sulfatoethyl sulfone, which alkali converts to the reactive vinyl form): cellulose−O^- adds across the double bond.

Bifunctional dyes carry two reactive groups (e.g. monochlorotriazine + vinyl sulfone) so that if one hydrolyses the other may still fix. The enemy reaction is **hydrolysis**: water competes with the fibre for the reactive group, and hydrolysed dye can never fix — it must be washed off or wash fastness fails. Fixation efficiency is therefore the class's central number:

WORKED EXAMPLE 1 — FIXATION EFFICIENCY AND EFFLUENT DYE

A reactive dye applied at 3.0% owf exhausts 78% onto cotton; of the exhausted dye, 85% reacts and fixes (the rest hydrolyses on the fibre or in the bath and is washed off).

Overall fixation = 0.78 × 0.85 = **66.3%** of applied dye.

Unfixed dye = 3.0% × (1 − 0.663) = **1.01% owf** → on a 1,000 kg production day, **10.1 kg of dye** leaves in wash water. This is why high-fixation dye ranges, low-salt systems and wash-off efficiency dominate reactive-dye development — the colour in the effluent is bought dye.

Molecular design notes: reactive dyes are mostly azo (yellows–reds–blacks) and anthraquinone or phthalocyanine (bright blues and greens); metal-complex azos serve deep navies and blacks. Solubilising sulfonate groups are balanced against substantivity — the dye must be substantive enough to exhaust (Topic 1, Freundlich behaviour on cellulose) yet soluble enough to apply levelly.

3. Disperse dyes — colour by solid solution

Disperse dyes are small, non-ionic, sparingly soluble molecules (azo and anthraquinone mainly, plus methine and benzodifuranone types) applied as a fine aqueous **dispersion** stabilised by dispersing agents. They colour hydrophobic fibres — polyester above all, also acetate and nylon — by dissolving into the fibre as a solid solution (Nernst partition, Topic 1). Above the polyester glass transition (~80 °C in water) the polymer chains gain mobility and admit the dye; production uses 130 °C HTHP, carriers (largely historical, for environmental reasons), or dry-heat thermosol fixation. Sublimation fastness is the class's molecular design constraint: small molecules dye fast and level well but sublime; larger molecules fix better but diffuse slowly — the energy/type classification (low, medium, high energy) organises the trade-off.

4. Azoic (naphthol) dyes — colour synthesised inside the fibre

Azoic colour is not applied; it is **manufactured in situ**. The fibre is first impregnated with a coupling component (a Naphthol AS derivative) dissolved as its sodium salt, then treated with a diazotised base (Fast Colour Base/Salt). Inside the fibre the diazonium ion couples with the naphthol to form an insoluble azo pigment mechanically trapped in the fibre — brilliant, wash-fast reds, oranges, bordeaux and blacks at low cost, historically the colour of choice for bright fast reds on cotton. The chemistry demands control: diazotisation needs cold acidic conditions with nitrous acid (excess nitrite destroyed before coupling), and naphtholated goods must be buffered correctly or coupling is incomplete and rub fastness suffers (pigment formed on the surface rather than within). Effluent and handling of aromatic amine chemistry are the class's modern constraints.

5. Sulphur dyes — colour by polymeric reduction–oxidation

Sulphur dyes are complex polymeric structures containing sulfur linkages, made by heating aromatic intermediates with sulfur or polysulfides. They are insoluble in water; dyeing reduces them with sodium sulfide (or modern reducing systems) to soluble **leuco** forms substantive to cellulose, then oxidises the dye back to the insoluble pigment inside the fibre. The class gives economical deep shades — blacks, navies, browns, olives — with good wash fastness on cotton, at the cost of dull hues, chlorine sensitivity, and a historic effluent burden of sulfide that modern pre-reduced and low-sulfide products address. Storage tendering of sulphur-dyed cotton by acid formation is a known fault to guard against.

6. Pigments — colour without affinity

Pigments are insoluble coloured particles with **no affinity for any fibre**. They are applied in a dispersion with a polymeric **binder**, padded or printed on, dried and cured; the binder film glues the pigment to the fabric surface. Consequences follow directly from the mechanism: any fibre or blend can be coloured by the same system; there is no wash-off stage and very little effluent colour; but the film sits on the surface, so rub fastness and handle depend entirely on binder quality and curing, and deep shades with high pigment loads feel harsh. Pigment systems dominate printing (WP-4101 Topic 4) and are growing in pad dyeing for exactly these environmental reasons.

7. The five systems compared

CLASS	COLOURED SPECIES	ATTACHMENT MECHANISM	PRINCIPAL FIBRES	SIGNATURE STRENGTH / WEAKNESS
Reactive	Azo, anthraquinone, phthalocyanine dyes	Covalent bond to fibre	Cotton, cellulosics, wool, silk, nylon	Brilliant + wash-fast / hydrolysis loss, salt load
Disperse	Small non-ionic azo/AQ molecules	Solid solution (partition)	Polyester, acetate, nylon	Only route for polyester / sublimation limits design
Azoic	Azo pigment formed in fibre	Insoluble pigment trapped in situ	Cotton	Brilliant fast reds, cheap / two-stage chemistry, rub care
Sulphur	Polymeric S-containing structures	Leuco adsorption, oxidised in fibre	Cotton	Economical deep shades / dull hues,

				sulfide effluent
Pigment	Insoluble coloured particles	Binder film on surface	All fibres and blends	Simple, low effluent / rub fastness and handle limits

8. Glossary — key terms

- **Chromophore:** the conjugated system whose electronic transitions produce colour.
- **Auxochrome:** a substituent modifying colour and assisting attachment.
- **Hydrolysis (of reactive dyes):** reaction of the reactive group with water, destroying fixability.
- **Leuco form:** the reduced, soluble form of a dye (sulphur, vat) with substantivity for the fibre.
- **Diazo coupling:** reaction of a diazonium salt with a coupling component to form an azo compound.
- **Binder:** the film-forming polymer that fixes pigment to the fabric surface.

9. Key points

- Classify a colourant by its attachment mechanism and the rest of its behaviour — fastness, effluent, faults — follows.
- Reactive fixation is a competition between fibre and water: 66.3% overall fixation leaves 10.1 kg of dye per tonne of fabric for the drain.
- Pigments break the rules of dyeing (no affinity, no wash-off) and pay for it in rub fastness and handle.

10. Summary

Colour chemistry is molecular engineering with a purpose: the triazine ring and the vinyl sulfone exist to make covalent bonds; the disperse dye's small non-ionic body exists to dissolve in polyester; the azoic pigment exists only where it is born, inside the fibre. Knowing which mechanism is at work tells the technologist which variables — alkali, temperature, reduction, binder cure — will decide success.

11. Practice

1. A reactive dye exhausts 82% and fixes 80% of what exhausts. Calculate overall fixation and the unfixed dye (kg) when 25 kg of dye is applied in a day.
2. Why must azoic coupling be carried out with the naphthol inside the fibre before the diazo component is applied — what goes wrong if both are simply mixed in one bath?

3. A buyer demands the same bright red on a cotton fabric and on a polyester fabric. Which class from this sheet serves each, and what different fixation conditions does each demand?
4. Explain, from mechanism, why pigment coloration needs no wash-off stage but is limited in rub fastness.

Answer key

1. Overall fixation = 0.82 × 0.80 = 65.6%. Unfixed = 25 × (1 − 0.656) = 8.6 kg.
2. Coupling in the bath would form the insoluble pigment outside the fibre, where it can only deposit on the surface and rub off; the colour would also be wasted in the liquor. Impregnating the fibre with naphthol first ensures the diazonium diffuses in and the pigment forms within the fibre structure, where it is mechanically held.
3. Cotton: reactive dyes (alkali fixation, typically soda ash at 50–80 °C exhaust or pad processes) — azoic is the classic alternative for this hue. Polyester: disperse dyes at 130 °C under pressure (HTHP) or thermosol dry heat, because only small non-ionic molecules dissolve into polyester.
4. Pigment is never dissolved or reacted, so there is no unfixed dye in a bath to remove — the application liquor is a dispersion, and what is padded on is cured in place. But attachment is only a surface film: abrasion physically removes binder and pigment, so rubbing fastness is set by film properties, not by any bond to the fibre.

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